

R18 Runtime Cascade

Packet window: 20 x 17 features

Tier-1: Random Forest Gate

Fast allow if $P(\text{BENIGN}) \geq 0.90$
156 KB model, 3.66 ms avg

Benign?

ESCALATE

ALLOW

Tier-2: CNN-GRU (ONNX)

Six classes including BENIGN
Block > 0.95 , flag $0.80-0.95$
0.18 ms avg ONNX inference

Tier-3: Mahalanobis Anomaly

128-D embedding, zero-day flag

Final action: ALLOW, FLAG, or BLOCK